



Department of Artificial Intelligence and Data Science

Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch: B.Tech , V & Artificial Intelligence and Data Science

Course Code & Title: CCS334 & Big Data Analytics

Name of the Faculty member (s): Dr.R.Ramana

Innovative Practice Description

- **Unit / Topic:** Unit II /Cassandra and its data model with examples
- **Course Outcome:** CO2
- **Topic Learning Outcome:** LO12
- **Activity Chosen:** Tool Exploration Challenge
- **Justification:**
 - ❖ A Tool Exploration Challenge was conducted to assess students' practical understanding of Cassandra and its data model. This activity reinforced their knowledge of NoSQL concepts, focusing on real-time data modeling and CQL operations used in Big Data systems.
- **Time Allotted for the Activity:** 45 Minutes
- **Details of the Implementation:** The Tool Exploration Challenge was designed to promote practical learning through direct interaction with the Cassandra environment.
 - ❖ **Step 1: Introduction and Setup**
 1. Students were introduced to the basics of Apache Cassandra architecture—peer-to-peer structure, replication, and partitioning.
 2. They installed and configured Cassandra on their systems or accessed an online Cassandra sandbox environment.
 - ❖ **Step 2: Exploration Tasks**
 - ❖ Each student explored and performed the following tasks:
 1. Create a keyspace for a healthcare or retail dataset.
 2. Define tables using primary key and clustering columns.
 3. Insert and retrieve sample data using CQL (Cassandra Query Language).
 4. Demonstrate data replication and consistency levels using simple queries.
 - ❖ **Step 3: Challenge Phase**
 1. Students were given a problem scenario—for example, designing a Cassandra data model to store and query patient health records or online shopping transactions.
 2. They submitted their CQL scripts and explained how their data model ensures scalability and availability.
 - ❖ **Step 4: Presentation and Discussion**
 1. Selected students presented their exploration findings, challenges, and observations.
 2. Faculty guided a discussion on the differences between Cassandra's data model and traditional RDBMS design.



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- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- Students appreciated the hands-on experience with a real NoSQL tool.
- The challenge improved their practical understanding of distributed databases and data modeling concepts.
- They gained confidence in using CQL commands to handle real-time datasets.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- Install and navigate the Cassandra environment.
- Understand and implement key concepts of Cassandra's data model.
- Write and execute basic CQL commands for data creation, insertion, and querying.
- Analyze how Cassandra ensures scalability, fault tolerance, and high availability.
- Relate NoSQL data management to real-world Big Data applications.

- ❖ **Challenges faced in implementation:**

- Some students faced installation issues due to system compatibility.
- Understanding Cassandra's distributed architecture and replication concepts required additional guidance.
- Writing complex CQL queries for nested data structures was challenging for beginners.
- Limited lab time restricted full exploration of advanced Cassandra features.



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FEEDBACK

Active Learning Best practices: Tool Exploration Challenge

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Course Code & Title : CCS334& Big Data Analytics

Name of the Faculty member (s) : Dr. R. Ramana, AP / AD

Theme of discussion: Cassandra and its data model with examples

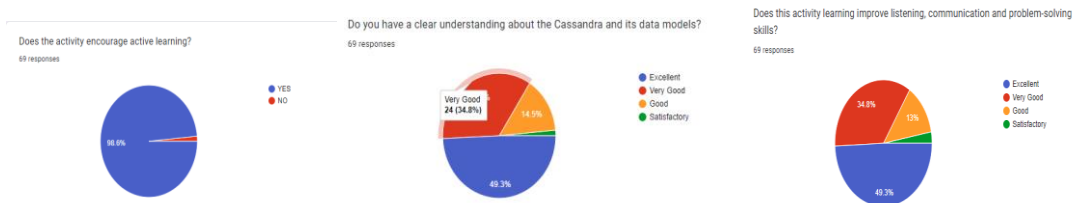
Date and Time: 23.08.2025 & 11.35 am to 12.20 pm

Feedback collected in class and also through online

Feedback Questionnaire:

- Does the activity encourage active learning? **Yes No**
- Do you have a clear understanding about the Cassandra and its data models?
Excellent Very Good Good Satisfactory
- Does this activity learning improve listening, communication and problem-solving skills?
Excellent Very Good Good Satisfactory

Feedback Analysis:



R.P.

Faculty in-charge

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HOD/AD